

Personal Projects

🌟 Self-initiated projects I built to learn and explore — distinct from company/freelance work. (Company work such as **iTax**, **Al-Zira**, **Sakhi**, **ESG Ratings** and **Mark AI** is intentionally kept under **Professional Experience**, even where its code sits on my personal GitHub.)

Projects at a glance

Project	What it is	Stack
Loupe	Screen recorder and video editor for Mac and Windows	Electron, Swift, .NET 8, WebCodecs, Whisper, ONNX
Souffleur	Screen-share-invisible AI assistant for calls	Electron, Swift, Gemini / OpenAI, pdf.js
SessionVault	Browser extension that exports a logged-in session	Manifest V3, Next.js
Cosmic-Trek	3D solar-system simulator with a RAG chatbot	Three.js, WebGL, D3.js, RAG
UpInTheSky	Travel-booking platform frontend	React, Firebase, Tailwind CSS
Legal E-Library	Digital library for legal case law	Angular / Next.js, Express, Prisma, PostgreSQL
SipCraft	E-commerce site with no frameworks	HTML, CSS, JavaScript, Stripe

Loupe — Screen Recorder & Video Editor (Flagship)

Aug 2026 – Present · Electron · Swift · .NET 8 · WebCodecs · Whisper · ONNX Runtime —
github.com/akshatgg/loupe · loupe.akshatgg.in

A desktop screen recorder for macOS and Windows that zooms in while you record, made for demo and tutorial videos where the viewer has to see the detail. 182 commits, 5 public releases (latest v2.1.1).

- **Zoom while recording.** Hold a modifier key (or a mouse side button) and scroll to zoom up to 4×; the view follows the cursor smoothly. While zoomed, a frame on screen shows exactly what the video will show — and that frame and the control bar are never recorded.
- **Record** the whole screen, one window, or a dragged rectangle inside a window.
- **Native capture written twice behind one protocol.** On macOS it is Swift (ScreenCaptureKit, CGEventTap, AVFoundation); on Windows it is one .NET 8 program (Windows.Graphics.Capture, low-level input hooks, Media Foundation). The Electron main process drives the same four helpers on both platforms: list sources, capture, input tap and render.
- **Non-linear editor.** A timeline with cut, trim and undo history; jump to, remove or restore zooms; speed ramps from 0.25× to 8× using a WSOLA time-stretch that keeps voices at a natural pitch; annotations, webcam bubble, keystroke overlay, wallpaper backgrounds and transitions composited as layers.
- **On-device captions.** Whisper through transformers.js and ONNX Runtime in a worker, with word-level timings, automatic language detection and SRT/VTT export.
- **Audio chain.** RNNoise denoising, music ducking and voiceover recording.

- **Export pipeline.** MP4, WebM and GIF at 1080p, 1440p or 4K, 60 fps, with bitrate planning that hits either a quality target or a hard file-size limit.
- **Shipping.** A DMG (Apple Silicon and Intel), a Homebrew cask (`brew install --cask akshatgg/tap/loupe`) and a Windows installer, built by a release pipeline.
- **Private by design.** Everything runs on the computer: no account, no upload.
- **Tested.** 60+ unit test files and Electron end-to-end suites.

Souffleur — Invisible AI Assistant for Calls

Sep 2026 – Present · Electron · Swift (ScreenCaptureKit) · Gemini / OpenAI · pdf.js · Vercel · Firebase — getsouffleur.vercel.app

A small always-on-top window that sees your screen, hears the call and answers questions, while staying out of the screen-share stream that everyone else is watching. Named after the theatre prompter hidden below the stage. 6 releases for macOS and Windows.

- **Hidden from screen capture.** The overlay does not appear in Zoom, Meet, Teams, OBS or QuickTime. I verified this by measurement rather than by reading docs: three test windows captured through a live 30 fps ScreenCaptureKit stream, 151 frames analysed, the protected window hidden in every frame and the unprotected control visible in every frame.
- **Hears the call.** Transcribes what the other party says straight into the chat box. Windows uses a loopback device; macOS has none, so I wrote a small Swift ScreenCaptureKit audio helper. Both feed the same silence gate and transcription.
- **Ask about your screen.** One hotkey captures the display the overlay is on and sends it with the question.
- **Interview mode.** Upload a CV and project documents as PDFs; text is extracted once with pdf.js (pure JavaScript, so it works on every platform) and the assistant answers in the first person from that history.
- **Any model.** Gemini, OpenAI, or any OpenAI-compatible endpoint (Groq, OpenRouter, Ollama and others), all streaming, with a hand-written SSE parser and live model listing from the user's own key.
- **Free key.** Works with no API key through a small rate-limited server (Vercel function, per-install and per-IP daily limits) that I can switch on and off from GitHub.
- **In-app updater.** An "Update now" button downloads the newest release, installs it over the running copy and reopens.
- **Shipping.** Homebrew cask, DMG and Windows installer from a GitHub Actions release pipeline; a marketing site with Google sign-up before downloads.

SessionVault — Session Export Extension

Jun 2026 · Manifest V3 (Chrome / Edge / Brave / Firefox) · Next.js — github.com/akshatgg/Sessionvault · sessionvault-pi.vercel.app

- A cross-browser extension that captures a tab's cookies and localStorage and exports them in Playwright `storageState` format, so a logged-in session can be reused in automated tests or another browser.
- Three outputs: copy to clipboard, download `storageState.json`, or POST to an endpoint you control with your own API key.

- No server and no database: the extension stores nothing, and the capture exists only in that one request.
- A Next.js documentation site with example receivers in Express and FastAPI.

Cosmic-Trek (Orrey App)

Dec 2025 – Mar 2026 · Three.js · WebGL · GSAP · Vite · D3.js · RAG —

github.com/akshatgg/OrreyApp

- Interactive 3D solar-system simulator with Three.js and WebGL — 8+ celestial bodies and 50+ exoplanets; about 60% faster rendering.
- Physics-based satellite-launch simulator with 5+ orbital patterns (about 30% lower computation load).
- Realistic collision and gravitational simulations with 10+ custom impact effects.
- A RAG chatbot grounded strictly in the Orbital Mechanics (Stanford) textbook — it answers only from that source and declines out-of-scope questions.
- Built as modular vanilla JS (`solar_system.js`, `planet.js`, `gravity_field.js`, `satellite_class.js`, `trail.js`) with GSAP animations, dat.gui controls and a Dockerised build.

UpInTheSky

Aug – Oct 2025 · React · Firebase · Firestore · Tailwind CSS · JavaScript

- Built the complete React frontend and UI for a travel-booking platform — reusable components and a responsive Tailwind layout across every page.
- Implemented the booking experience: browse curated packages, view destination details and complete an end-to-end booking.
- Integrated Firebase — Firestore for real-time data and Firebase Auth for secure login — serving 100+ travel listings.

Legal E-Library

Mar – Jun 2025 · Angular / Next.js · Node.js · Express.js · Prisma ORM · PostgreSQL —

github.com/akshatgg/E_Library

- Built a digital library for legal case laws with advanced search and filtering — about 40% better research efficiency.
- Responsive UI giving access to 10,000+ legal documents across 50+ categories.
- Backend REST API with Node.js, Express and Prisma ORM over PostgreSQL, with about 60% query-performance optimisation.

SipCraft

Vanilla HTML · CSS · JavaScript · Stripe — github.com/akshatgg/SipCrafts

- Built a complete e-commerce website from scratch in vanilla HTML/CSS/JS, with no frameworks: a responsive product catalog and storefront.
- Implemented the full shopping flow in plain JavaScript: add-to-cart, a dynamic cart with live totals, and checkout.
- Integrated Stripe for secure end-to-end online payments.

More on GitHub

Beyond the projects above, my @akshatgg account includes a range of learning and exploration work:

- **Small sites** — Yamoto, a statically generated Next.js 16 landing page for a wheel and tyre shop with local SEO (JSON-LD structured data, sitemap, Lighthouse 100 for SEO, accessibility and best practices).
- **3D and creative** — a Three.js 3D developer portfolio, particle and animation experiments.
- **DSA and fundamentals** — `DSA_By_JAVA`, `LeetCodeSoln`, C/C++ programming sets.
- **Web apps** — News App, Grocery store, LMS portals, several React/Next.js portfolios and dashboards.
- **Explorations** — `fastner` (TypeScript + FastAPI), `new_portfolio`, and a Homebrew tap that distributes Loupe and Souffleur.

Profile: github.com/akshatgg